

EVALUATION-2 — PRACTICE MCQ PAPER

Topics	Ratio & Partnership Speed/Time/Distance & Trains Blood Relations Direction Sense 5
Questions	25 Questions 1 Mark each Total: 25 Marks
Duration	45 Minutes (Practice target: complete this paper within 45 minutes)
Difficulty	Moderate — similar to Evaluation-2 exam pattern

SECTION A: RATIO & PARTNERSHIP [Q1–Q5]

1. The ratio of ages of A and B is 4:5. After 10 years their ratio will be 6:7. Find A's present age.

- A) 20 years
- B) 25 years
- C) 30 years
- D) 35 years

2. Rs. 6,500 is divided among A, B, C in ratio $\frac{1}{2} : \frac{1}{3} : \frac{1}{4}$. Find B's share.

- A) Rs. 1,500
- B) Rs. 2,000
- C) Rs. 1,800
- D) Rs. 2,500

3. A, B, C invest in 3:4:5 ratio. Total profit is Rs. 48,000. Find A's profit.

- A) Rs. 12,000
- B) Rs. 16,000
- C) Rs. 14,400
- D) Rs. 9,600

4. A and B enter a partnership. A invests Rs. 10,000 for 8 months and B invests Rs. 12,000 for 6 months. Total profit is Rs. 19,000. B's share?

- A) Rs. 9,000
- B) Rs. 10,000
- C) Rs. 8,000
- D) Rs. 11,000

5. If $A:B = 3:4$ and $B:C = 8:9$, find $A:B:C$.

- A) 6:8:9
- B) 3:8:9
- C) 24:32:36
- D) 3:4:9

SECTION B: SPEED, TIME & DISTANCE AND TRAINS [Q6–Q10]

6. A train travels at 72 km/h. Convert to m/s.

- A) 20 m/s
- B) 18 m/s
- C) 25 m/s
- D) 15 m/s

7. A train 200 m long crosses a bridge 300 m long in 25 seconds. Speed of train?

- A) 60 km/h
- B) 72 km/h
- C) 54 km/h
- D) 80 km/h

8. Two trains 150 m and 100 m run in opposite directions at 60 km/h and 40 km/h. Time to cross?

- A) 9 sec
- B) 10 sec
- C) 12 sec
- D) 8 sec

9. Ravi travels from home to office at 30 km/h and returns at 20 km/h. Average speed?

- A) 24 km/h
- B) 25 km/h
- C) 22 km/h
- D) 26 km/h

10. A man walks from X to Y at 5 km/h and back at 3 km/h, taking 4 hours total. Distance X to Y?

- A) 7.5 km
- B) 6 km
- C) 8 km
- D) 10 km

SECTION C: BLOOD RELATIONS [Q11–Q15]

11. Pointing to a man, Priya says 'His mother is the only daughter of my mother.' How is Priya related to the man?

- A) Aunt
- B) Sister
- C) Mother
- D) Grandmother

12. A is the brother of B. B is the sister of C. C is the son of D. How is D related to A?

- A) Father
- B) Mother
- C) Uncle
- D) Parent (father or mother)

13. Rahul says, 'Seema's father is the only son of my grandfather.' How is Rahul related to Seema?

- A) Father
- B) Brother
- C) Uncle
- D) Cousin

14. In a family: M is father of N, N is sister of O, O is son of P, P is wife of M. How is M related to O?

- A) Grandfather
- B) Uncle
- C) Father
- D) Brother

15. Coded relations: $A + B$ means A is husband of B; $A - B$ means A is wife of B; $A \times B$ means A is father of B; $A \div B$ means A is daughter of B. In $R \times S + T \div U$, what is R's relation to U?

- A) Father-in-law
- B) Grandfather
- C) Son-in-law
- D) Uncle

SECTION D: DIRECTION SENSE [Q16–Q20]

16. A starts from O, walks 3 km East, turns left, walks 4 km. Shortest distance from O?
- A) 7 km
 - B) 5 km
 - C) 6 km
 - D) 4 km
17. Mohan walks 10 km South, turns right, walks 5 km, turns right, walks 10 km. Distance from start?
- A) 25 km
 - B) 10 km
 - C) 5 km
 - D) 15 km
18. In morning, a man stands facing his shadow. Which direction is he facing?
- A) East
 - B) West
 - C) North
 - D) South
19. Sita goes 5 km North, then 12 km East, then 5 km South. How far from start and in which direction?
- A) 12 km East
 - B) 13 km NE
 - C) 10 km East
 - D) 17 km NE
20. A faces North. She turns 90° clockwise, then 45° anticlockwise. Which direction is she facing?
- A) North-East
 - B) East
 - C) South-East
 - D) North

SECTION E: SYLLOGISM [Q21–Q25]

21. Statements: All pens are caps. All caps are books.
Conclusions: I. All pens are books. II. All books are pens.
- A) Only I
 - B) Only II
 - C) Both
 - D) Neither
22. Statements: No cat is a rat. Some rats are bats.
Conclusions: I. Some bats are not cats. II. No bat is a cat.

- A) Only I
- B) Only II
- C) Both
- D) Neither

23. Statements: Some tables are chairs. No chair is wooden.

Conclusions: I. Some tables are not wooden. II. All tables are wooden.

- A) Only I
- B) Only II
- C) Both
- D) Either I or II

24. Statements: All mangoes are sweet. Some sweets are healthy.

Conclusions: I. Some mangoes are healthy. II. All healthy things are mangoes.

- A) Both
- B) Only I
- C) Only II
- D) Neither

25. Statements: No star is a moon. No moon is a planet.

Conclusions: I. No star is a planet. II. Some planets are stars.

- A) Only I
- B) Only II
- C) Both
- D) Either I or II

ANSWER KEY & DETAILED SOLUTIONS

Q#	Answer	Q#	Answer	Q#	Answer	Q#	Answer	Q#	Answer
1	A)	2	B)	3	A)	4	A)	5	A)
6	A)	7	B)	8	A)	9	A)	10	A)
11	C)	12	D)	13	A)	14	C)	15	A)
16	B)	17	C)	18	B)	19	A)	20	A)
21	A)	22	A)	23	A)	24	D)	25	D)

SECTION A: RATIO & PARTNERSHIP

Q1. 1. The ratio of ages of A and B is 4:5. After 10 years their ratio will be 6:7. Find A's present age.

✓ **A) 20 years**

→ $4x$ and $5x$. $(4x+10)/(5x+10)=6/7 \rightarrow 28x+70=30x+60 \rightarrow 2x=10 \rightarrow x=5$. $A=4 \times 5=20$ years.

Q2. 2. Rs. 6,500 is divided among A, B, C in ratio $1/2 : 1/3 : 1/4$. Find B's share.

✓ **B) Rs. 2,000**

→ Ratios: $1/2 : 1/3 : 1/4 = 6:4:3$ (multiply by LCM=12). Sum=13. $B=(4/13) \times 6500 = \text{Rs. } 2,000$.

Q3. 3. A, B, C invest in 3:4:5 ratio. Total profit is Rs. 48,000. Find A's profit.

✓ **A) Rs. 12,000**

→ $A = (3/12) \times 48,000 = \text{Rs. } 12,000$

Q4. 4. A and B enter a partnership. A invests Rs. 10,000 for 8 months and B invests Rs. 12,000 for 6 months. Total profit is Rs. 19,000. B's share?

✓ **A) Rs. 9,000**

→ $A = 10000 \times 8 = 80,000$; $B = 12000 \times 6 = 72,000$. Ratio=80:72=10:9. $B=(9/19) \times 19,000 = \text{Rs. } 9,000$.

Q5. 5. If $A:B = 3:4$ and $B:C = 8:9$, find $A:B:C$.

✓ **A) 6:8:9**

→ $A:B=3:4=6:8$, $B:C=8:9$. So $A:B:C=6:8:9$.

SECTION B: SPEED, TIME & DISTANCE

Q6. 6. A train travels at 72 km/h. Convert to m/s.

✓ **A) 20 m/s**

→ $72 \times 5/18 = 20$ m/s

Q7. 7. A train 200 m long crosses a bridge 300 m long in 25 seconds. Speed of train?

✓ **B) 72 km/h**

→ Distance = $200+300=500$ m. Speed= $500/25=20$ m/s= 72 km/h.

Q8. 8. Two trains 150 m and 100 m run in opposite directions at 60 km/h and 40 km/h. Time to cross?

✓ **A) 9 sec**

→ Relative speed= 100 km/h= $250/9$ m/s. Total length= 250 m. Time= $250/(250/9)=9$ sec.

Q9. 9. Ravi travels from home to office at 30 km/h and returns at 20 km/h. Average speed?

✓ **A) 24 km/h**

→ Same distance: $\text{Avg} = \frac{2 \times 30 \times 20}{(30+20)} = \frac{1200}{50} = 24 \text{ km/h}$.

Q10. 10. A man walks from X to Y at 5 km/h and back at 3 km/h, taking 4 hours total. Distance X to Y?

✓ **A) 7.5 km**

→ $\frac{D}{5} + \frac{D}{3} = 4 \rightarrow 3D + 5D = 60 \rightarrow 8D = 60 \rightarrow D = 7.5 \text{ km}$.

SECTION C: BLOOD RELATIONS

Q11. 11. Pointing to a man, Priya says 'His mother is the only daughter of my mother.' How is Priya related to the man?

✓ **C) Mother**

→ Only daughter of Priya's mother = Priya herself. So Priya is the man's mother.

Q12. 12. A is the brother of B. B is the sister of C. C is the son of D. How is D related to A?

✓ **D) Parent (father or mother)**

→ C is D's son → D is C's parent. A, B, C are siblings → D is A's parent. Gender of D not stated = Parent.

Q13. 13. Rahul says, 'Seema's father is the only son of my grandfather.' How is Rahul related to Seema?

✓ **A) Father**

→ Rahul's grandfather's only son = Rahul's father. Seema's father = Rahul's father. → Rahul is Seema's brother. But 'only son' means Rahul IS that son. Seema's father = Rahul → Rahul is Seema's father.

Q14. 14. In a family: M is father of N, N is sister of O, O is son of P, P is wife of M. How is M related to O?

✓ **C) Father**

→ M is father of N. P is wife of M. O is son of P. M and P are married → M is also father of O.

Q15. 15. Coded relations: A + B means A is husband of B; A – B means A is wife of B; A × B means A is father of B; A ÷ B means A is daughter of B. In $R \times S + T \div U$, what is R's relation to U?

✓ **A) Father-in-law**

→ $R \times S$: R is father of S. $S + T$: S is husband of T. $T \div U$: T is daughter of U. R is father of S who is husband of T who is daughter of U. R is father of T's husband → R is T's father-in-law. But R is father of S, S is husband of T. R is S's father = T's father-in-law. Answer A.

SECTION D: DIRECTION SENSE

Q16. 16. A starts from O, walks 3 km East, turns left, walks 4 km. Shortest distance from O?

✓ **B) 5 km**

→ 3 East, 4 North. Distance = $\sqrt{(3+16)} = \sqrt{25} = 5 \text{ km}$ (3-4-5 triplet).

Q17. 17. Mohan walks 10 km South, turns right, walks 5 km, turns right, walks 10 km. Distance from start?

✓ **C) 5 km**

→ 10 S → (0,-10). Turn right=West, 5 km → (-5,-10). Turn right=North, 10 km → (-5,0). Net: 5 km West. Distance=5 km.

Q18. 18. In morning, a man stands facing his shadow. Which direction is he facing?

✓ **B) West**

→ Morning sun in East → shadow falls West. To face his shadow he must face West.

Q19. 19. Sita goes 5 km North, then 12 km East, then 5 km South. How far from start and in which direction?

✓ **A) 12 km East**

→ Net NS: 5N - 5S = 0. Net EW: 12 East. She is 12 km East of start.

Q20. 20. A faces North. She turns 90° clockwise, then 45° anticlockwise. Which direction is she facing?

✓ **A) North-East**

→ North + 90° CW = East. East - 45° (ACW) = North-East.

SECTION E: SYLLOGISM

Q21. 21. Statements: All pens are caps. All caps are books. Conclusions: I. All pens are books. II. All books are pens.

✓ **A) Only I**

→ A+A→A: All pens are books ✓. All books are pens — not derivable (too strong). Only I.

Q22. 22. Statements: No cat is a rat. Some rats are bats. Conclusions: I. Some bats are not cats. II. No bat is a cat.

✓ **A) Only I**

→ E+I→O: Some bats are not cats (I ✓). No bat is a cat — too strong, not derivable. Only I.

Q23. 23. Statements: Some tables are chairs. No chair is wooden. Conclusions: I. Some tables are not wooden. II. All tables are wooden.

✓ **A) Only I**

→ I+E→O: Some tables are not wooden (I ✓). Conclusion II is opposite and false. Only I.

Q24. 24. Statements: All mangoes are sweet. Some sweets are healthy. Conclusions: I. Some mangoes are healthy. II. All healthy things are mangoes.

✓ **D) Neither**

→ A+I (reverse middle term). The 'some sweets' that are healthy may or may not include mangoes. Neither follows definitively.

Q25. 25. Statements: No star is a moon. No moon is a planet. Conclusions: I. No star is a planet. II. Some planets are stars.

✓ **D) Either I or II**

→ E+E→No definite conclusion. Neither I nor II follows. But they form a complementary pair (E and I for same star-planet): Either I or II follows.